

HCSM-28

Orbit Class Reconstruction

The Winding Class Partition as a Reconstruction of the Dihedral Orbit Partition,
and the Decomposition of the Distinguished Pair Graph
The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model assigns the fourteen modes at the self paired eigenvalue of the discrete Laplacian on the periodic eight by eight torus to one Standard Model generation. On this multiplet, three independent partitions act: the dihedral orbit partition of sizes two, eight, and four; the winding class partition of sizes three, seven, and four; and the fermion/boson split of sizes ten and four. This paper proves two theorems that characterize the relationship between these partitions and the pair structure of the multiplet. First, the winding class partition is the dihedral orbit partition modified by a unique two mode swap: the top quark moves from the size four dihedral orbit into winding class A, and the dark photon moves from the size eight dihedral orbit into winding class C. This swap is uniquely forced by requiring each winding class to be fermion/boson homogeneous. Second, the graph of the ten distinguished pairs of the multiplet decomposes into four isolated two vertex edges and one connected six vertex core. Every isolated edge is homogeneous in dihedral orbit, winding class, and fermion/boson content, and heterogeneous in charge line sign. The six vertex core is heterogeneous in every structural feature, has a unique three degree hub at the dark photon, and contains exactly one independent cycle. Both theorems are HCSM-native, and all numerical claims are verified at machine precision.

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1 Introduction

1.1 The pair structure of the fourteen mode multiplet

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic eight by eight torus [1]. The fourteen modes at the self paired eigenvalue of the discrete Laplacian are identified with one Standard Model generation.

Three independent partitions organize the fourteen-mode multiplet.

The dihedral orbit partition.

The fourteen modes decompose under the dihedral point group of the square into three orbits of sizes two, eight, and four [2].

The winding class partition.

The fourteen modes partition into three classes by the frequency of the winding generator, of sizes three, seven, and four [4].

The fermion/boson split.

The processor partitions the fourteen modes into a fermion sector of size ten and a boson sector of size four [3].

This paper proves two theorems that characterize how these partitions relate, and how the pair structure of the multiplet decomposes.

1.2 The one-line summary

The winding class partition is the dihedral orbit partition plus a single swap of two modes. The swap is the minimal correction that makes each winding class fermion/boson homogeneous. The ten distinguished pairs of the multiplet then organize into four homogeneous edges and one connected core, with the core being the framework’s structural hub.

1.3 The two theorems of this paper

The paper is organized around two theorems.

1. **Orbit-Class Reconstruction Theorem** (Section 3). The winding class partition is the dihedral orbit partition modified by the unique two mode swap consisting of the top quark and the dark photon: the top quark moves from dihedral orbit two into winding class A, and the dark photon moves from dihedral orbit one into winding class C. The swap is uniquely forced by requiring each winding class to be fermion/boson homogeneous.
2. **Distinguished Pair Graph Decomposition Theorem** (Section 4). The graph of the ten distinguished pairs of the fourteen-mode multiplet decomposes into four isolated two-vertex edges and one connected six vertex core. Every isolated edge is homogeneous in dihedral orbit, winding class, and fermion/boson content, and heterogeneous in charge line sign. The six vertex core is heterogeneous in every structural feature, has a unique three degree hub at the dark photon, and contains exactly one independent cycle.

1.4 Cross references to the HCSM paper stack

The paper depends on four companion papers in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-04** [2]: the fourteen-mode multiplet and its dihedral orbit decomposition.
- **HCSM-14** [4]: the winding generator spectrum and the winding class partition.
- **HCSM-24** [7]: the charge theorem, giving the charge-line signs of each mode.
- **HCSM-27** [9]: the Klein orbit structure, from which the swap pair and the distinguished-pair graph are drawn.

Additional references to HCSM-13 (the processor T) and HCSM-23 (the mass-ordering labels) appear where needed.

1.5 Organization

Section 2 states the setup: the fourteen-mode multiplet, the three partitions, the swap pair, and the distinguished-pair graph. Section 3 proves the Orbit-Class Reconstruction Theorem. Section 4 proves the Distinguished-Pair Graph Decomposition Theorem. Section 5 discusses the structural interpretation. Section 6 gives reproducibility notes. Section 7 concludes.

2 Setup

This section states the HCSM primitives on which the paper depends. All statements are HCSM-native; the reader is referred to HCSM-00 for the full foundations.

2.1 The fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [1]. The discrete five-point Laplacian has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (1)$$

The self paired eigenvalue $\lambda = 4$ has multiplicity fourteen. In the standard mode order used throughout HCSM (matching the zero state vector of HCSM-22 [5]), the fourteen modes are

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (2)$$

The mode coordinates in this order are:

$$\begin{aligned} \nu : (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \quad s : (3, 1), \quad \mu : (3, 7), \\ \text{dark} : (5, 7), \quad c : (7, 3), \quad \tau : (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \\ H : (6, 2), \quad t : (6, 6). \end{aligned} \quad (3)$$

2.2 The dihedral orbit partition

The dihedral orbit partition of the fourteen-mode multiplet is

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}, \quad (4)$$

with sizes 2, 8, and 4 respectively.

2.3 The winding class partition

The winding generator H_{wind} on the fourteen-mode multiplet has spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$. The fourteen modes partition into three classes by frequency:

$$A = \{\nu, e, t\}, \quad \omega_A = 2\sqrt{2}, \quad (5)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad \omega_B = 6\sqrt{2}, \quad (6)$$

$$C = \{\text{dark}, W, Z, H\}, \quad \omega_C = 0, \quad (7)$$

with sizes 3, 7, and 4 respectively. See HCSM-14 for the full derivation.

2.4 The fermion/boson split

The processor T is the rank-ten orthogonal projection onto the fermion sector [3]:

$$F = \{\nu, e, u, d, s, \mu, c, \tau, b, t\}, \quad |F| = 10, \quad (8)$$

$$B = \{\text{dark}, W, Z, H\}, \quad |B| = 4. \quad (9)$$

The class partition and the fermion/boson split are related by the exact correspondence $A \cup B = F$ and $C = B$, proved in HCSM-13.

2.5 The charge-line sign

The charge theorem of HCSM-24 assigns to each mode a charge-line sign

$$\sigma = \text{sign}(s) \in \{-1, 0, +1\}, \quad (10)$$

where s is the zero-state amplitude. The three charge lines are

$$\sigma = -1 : \{\nu, d, s, \text{dark}, c, Z\}, \quad (11)$$

$$\sigma = 0 : \{W, t\}, \quad (12)$$

$$\sigma = +1 : \{e, \mu, \tau, b, H, u\}. \quad (13)$$

See HCSM-24 for the full charge theorem.

2.6 The swap pair

The swap pair is the two element set consisting of the top quark and the dark photon. It is the pair whose exchange relates the dihedral orbit partition to the winding class partition, as stated in the Orbit Class Reconstruction Theorem of Section 3.

2.7 The distinguished-pair graph

Definition 2.1 (Distinguished-pair graph). The *distinguished pair graph* of the fourteen-mode multiplet has a vertex for each of the fourteen modes, and an edge for each two element subset that is distinguished by at least one of the following structural criteria:

1. the subset is fixed by an involution of the dihedral group,
2. the subset is a two cycle of the diagonal reflection rs ,
3. the subset is a two cycle of the half-period translation T_{half} ,
4. the subset is the orbit-class swap pair (this criterion is defined in Section 3 and in HCSM-26 [8]).

The distinguished-pair graph has ten edges. They are:

$$\{\nu, e\}, \quad \{u, s\}, \quad \{u, \text{dark}\}, \quad \{d, b\}, \quad \{\mu, c\}, \quad (14)$$

$$\{s, \tau\}, \quad \{\text{dark}, \tau\}, \quad \{\text{dark}, t\}, \quad \{Z, H\}, \quad \{W, t\}. \quad (15)$$

3 The Orbit Class Reconstruction Theorem

This section proves that the winding class partition is the dihedral orbit partition modified by a unique two-mode swap.

3.1 Comparison of the partitions

Lemma 3.1 (Orbit class differences). *The dihedral orbit partition $\{\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2\}$ and the winding class partition $\{A, B, C\}$ of the fourteen mode multiplet agree on twelve of the fourteen modes. They differ on exactly two modes:*

- the top quark t lies in dihedral orbit $\mathcal{O}_2$ but in winding class A ,
- the dark photon lies in dihedral orbit $\mathcal{O}_1$ but in winding class C .

Proof. Direct inspection of the two partitions.

Mode	Dihedral orbit	Winding class
ν	$\mathcal{O}_0$	A
e	$\mathcal{O}_0$	A
u	$\mathcal{O}_1$	B
d	$\mathcal{O}_1$	B
s	$\mathcal{O}_1$	B
μ	$\mathcal{O}_1$	B
c	$\mathcal{O}_1$	B
τ	$\mathcal{O}_1$	B
b	$\mathcal{O}_1$	B
W	$\mathcal{O}_2$	C
Z	$\mathcal{O}_2$	C
H	$\mathcal{O}_2$	C
dark	$\mathcal{O}_1$	C
t	$\mathcal{O}_2$	A

The twelve modes above the line agree on their dihedral orbit and winding class. The two modes below the line differ. $\square$

Corollary 3.2 (Swap formulation). *The winding class partition is obtained from the dihedral orbit partition by the two mode swap*

$$t : \mathcal{O}_2 \rightarrow A, \quad \text{dark} : \mathcal{O}_1 \rightarrow C. \quad (16)$$

Proof. Immediate from the lemma. $\square$

3.2 Uniqueness of the swap

Theorem 3.3 (Orbit Class Reconstruction Theorem). *The swap of Corollary 3.2 is the unique two-mode swap that makes every winding class fermion/boson-homogeneous. Equivalently, the winding class partition is the unique reconstruction of the dihedral orbit partition by a two-mode swap that yields fermion/boson homogeneity.*

Proof. We characterize the constraint imposed by fermion/boson homogeneity and solve.

Step 1: Without a swap, the classes are not homogeneous. The dihedral orbit $\mathcal{O}_2 = \{W, Z, H, t\}$ contains three bosons $\{W, Z, H\}$ and one fermion t . The dihedral orbit $\mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}$ contains one boson dark and seven fermions. The dihedral orbit $\mathcal{O}_0 = \{\nu, e\}$ is homogeneous (both fermions). Hence the dihedral orbit partition does not yield fermion/boson homogeneity for the winding classes.

Step 2: The swap must move t and dark. For the winding class partition to be homogeneous, class A (of size 3) must be either all-fermion or all-boson, and class C (of size 4) must be either all-fermion or all-boson, and class B (of size 7) must be either all-fermion or all-boson.

Since $\mathcal{O}_0 = \{\nu, e\}$ is homogeneous and class A contains ν and e , class A is already non-empty with two fermions. Class A has size 3 and needs one more mode. The candidate modes to add are the remaining twelve modes. For class A to be homogeneous, the added mode must be a fermion. This rules out adding any boson from $\{W, Z, H, \text{dark}\}$. The only way to keep class A homogeneous while also making class C homogeneous is to add the fermion t to class A and to move the boson dark into class C . This is the swap.

Step 3: The swap is unique. Suppose we attempt a different swap. If we add a different fermion (say, u) to class A , then class A remains homogeneous, but class C still contains $\{W, Z, H\}$ plus the mode u was removed from — which does not make class C homogeneous.

Similarly for any other fermion. If we add a boson to class A , class A is no longer homogeneous. If we move a boson out of class C (say, move W to class B) and move a fermion into class A , then class C still contains $\{Z, H, \text{dark}\}$ (three bosons) and needs to gain one boson to reach size 4, which requires a fourth mode — the swap is no longer two-mode. Hence the two-mode swap of t and dark is the unique one.

Step 4: Explicit check. With the swap, the three classes are

$$A = \{\nu, e, t\}, \quad \text{all fermions}, \quad (17)$$

$$B = \{u, d, s, \mu, c, \tau, b\}, \quad \text{all fermions}, \quad (18)$$

$$C = \{W, Z, H, \text{dark}\}, \quad \text{all bosons}. \quad (19)$$

Each class is fermion/boson-homogeneous. This is exactly the correspondence $A \cup B = F$ and $C = B$ proved in HCSM-13. $\square$

Corollary 3.4 (Structural interpretation). *The winding class partition is not an independent structure on the fourteen-mode multiplet. It is the dihedral orbit partition corrected by the minimal swap that makes the partition fermion/boson-homogeneous. The correction is uniquely determined by the dihedral orbit partition and the fermion/boson split.*

Proof. Immediate from the theorem. $\square$

Remark 3.5. The swap pair $\{\text{dark}, t\}$ is the same pair whose symmetric-function fingerprint (sum, difference, product, ratio) is entirely framework-constant, as proved in the Swap-Pair Fingerprint Theorem of HCSM-26 [8]. The two theorems are independent derivations of the same pair.

4 The Distinguished-Pair Graph Decomposition

This section proves that the distinguished-pair graph decomposes into four isolated edges and one connected six-vertex core.

4.1 The ten edges

The distinguished-pair graph has ten edges, listed in Section 2 and in HCSM-26. We now partition them into connected components.

4.2 Component decomposition

Theorem 4.1 (Distinguished-Pair Graph Decomposition Theorem). *The distinguished-pair graph decomposes into four isolated two-vertex edges and one connected six-vertex core:*

$$\{\nu, e\}, \quad \{d, b\}, \quad \{\mu, c\}, \quad \{Z, H\}, \quad \text{and} \quad \{u, s, \text{dark}, \tau, W, t\}. \quad (20)$$

Every isolated two-vertex edge is homogeneous in dihedral orbit, winding class, and fermion/boson content, and heterogeneous in charge-line sign. The six-vertex core is heterogeneous in every structural feature, has a unique three-degree hub at the dark photon, and contains exactly one independent cycle.

Proof. We proceed in three parts.

Part 1: Component decomposition. The ten edges are

$$\begin{aligned} &\{\nu, e\}, \quad \{u, s\}, \quad \{u, \text{dark}\}, \quad \{d, b\}, \quad \{\mu, c\}, \\ &\{s, \tau\}, \quad \{\text{dark}, \tau\}, \quad \{\text{dark}, t\}, \quad \{Z, H\}, \quad \{W, t\}. \end{aligned}$$

Consider the vertex ν . Its only incident edge is $\{\nu, e\}$, so ν is in a component with just e . Similarly, the components determined by the edges $\{d, b\}$ and $\{\mu, c\}$ and $\{Z, H\}$ are isolated edges: the vertices d, b and μ, c and Z, H have no other incident edges.

The remaining six edges are

$$\{u, s\}, \quad \{u, \text{dark}\}, \quad \{s, \tau\}, \quad \{\text{dark}, \tau\}, \quad \{\text{dark}, t\}, \quad \{W, t\}. \quad (21)$$

These six edges connect the six vertices $u, s, \text{dark}, \tau, t, W$ into a single connected component. The component decomposition is therefore as stated.

Part 2: Homogeneity of the isolated edges. Consider each isolated edge:

Edge	Dihedral orbit	Winding class	F/B content	Charge-line signs
$\{\nu, e\}$	$\mathcal{O}_0/\mathcal{O}_0$	A/A	F/F	$-1/+1$
$\{d, b\}$	$\mathcal{O}_1/\mathcal{O}_1$	B/B	F/F	$-1/+1$
$\{\mu, c\}$	$\mathcal{O}_1/\mathcal{O}_1$	B/B	F/F	$+1/-1$
$\{Z, H\}$	$\mathcal{O}_2/\mathcal{O}_2$	C/C	B/B	$-1/+1$

Each isolated edge is homogeneous in dihedral orbit, winding class, and fermion/boson content (both endpoints share each). Each is heterogeneous in charge-line sign (the two endpoints have opposite signs).

Part 3: Structure of the six-vertex core. The core is $\{u, s, \text{dark}, \tau, t, W\}$. Its vertices' structural features are:

Vertex	Dihedral orbit	Winding class	F/B	Charge-line sign
u	$\mathcal{O}_1$	B	F	$+1$
s	$\mathcal{O}_1$	B	F	-1
τ	$\mathcal{O}_1$	B	F	$+1$
dark	$\mathcal{O}_1$	C	B	-1
t	$\mathcal{O}_2$	A	F	0
W	$\mathcal{O}_2$	C	B	0

The core contains modes from both dihedral orbits $\mathcal{O}_1$ and $\mathcal{O}_2$, from all three winding classes A, B, C , from both fermion and boson sectors, and from all three charge-line signs. Hence the core is heterogeneous in every structural feature.

Degree distribution. The degrees of the six vertices in the core (counting only core edges) are:

$$\deg(\text{dark}) = 3 \quad (\text{edges } \{u, \text{dark}\}, \{\text{dark}, \tau\}, \{\text{dark}, t\}), \quad (22)$$

$$\deg(u) = 2 \quad (\text{edges } \{u, s\}, \{u, \text{dark}\}), \quad (23)$$

$$\deg(s) = 2 \quad (\text{edges } \{u, s\}, \{s, \tau\}), \quad (24)$$

$$\deg(\tau) = 2 \quad (\text{edges } \{s, \tau\}, \{\text{dark}, \tau\}), \quad (25)$$

$$\deg(t) = 2 \quad (\text{edges } \{\text{dark}, t\}, \{W, t\}), \quad (26)$$

$$\deg(W) = 1 \quad (\text{edge } \{W, t\}). \quad (27)$$

The dark photon has the unique maximum degree of 3.

Cycle rank. The core has $V = 6$ vertices and $E = 6$ edges. Its cycle rank is $E - V + 1 = 1$. There is exactly one independent cycle in the core, which is

$$\text{dark} \rightarrow u \rightarrow s \rightarrow \tau \rightarrow \text{dark}. \quad (28)$$

This is verified by direct traversal of the core edges. $\square$

Corollary 4.2 (Two-tier structure of the distinguished-pair graph). *The distinguished-pair graph has a two-tier structure: an outer shell of four isolated edges, each homogeneous in every partition, and an inner core of one connected six-vertex component, which is heterogeneous in every partition.*

Proof. Immediate from the theorem. □

Remark 4.3. The four isolated edges are exactly the four Klein orbits of the fourteen-mode multiplet with pointwise stabilizer generated by the product of the two generators, as proved in the Kernel Classification Theorem of HCSM-27. Each isolated edge is the union of the two members of one such Klein orbit, together with the edge connecting them.

The six-vertex core contains the two remaining Klein orbits: the observer orbit $\{W, t\}$ and the four-cycle orbit $\{\text{dark}, s, u, \tau\}$, together with all their internal edges and the bridge edge $\{\text{dark}, t\}$ between them. The core is the unique connected component that spans multiple Klein orbits.

5 Discussion

5.1 The reconstruction as a structural constraint

The Orbit-Class Reconstruction Theorem shows that the winding class partition is not an independent structure. It is the dihedral orbit partition, corrected by the unique two-mode swap that makes each winding class fermion/boson-homogeneous. This means that the winding generator and the processor are not independent inputs to the framework: the winding class structure is determined by the dihedral orbit structure together with the fermion/boson split.

The swap itself has a physical content: the top quark, a fermion, is moved from the boson-heavy orbit $\mathcal{O}_2$ into the fermion-rich class A ; the dark photon, a boson, is moved from the fermion-heavy orbit $\mathcal{O}_1$ into the boson class C . The swap aligns the fermion/boson content of the dihedral orbits with the fermion/boson content of the winding classes.

5.2 The two-tier graph structure

The Distinguished-Pair Graph Decomposition Theorem reveals a two-tier structure in the graph of distinguished pairs. The outer tier consists of four isolated edges, each one a self-contained unit that respects every partition of the multiplet. The inner tier consists of one connected core, which spans all three dihedral orbits, all three winding classes, both fermion and boson sectors, and all three charge-line signs.

The core's unique three-degree hub is the dark photon. The dark photon is the only mode whose degree exceeds 2. It is also the swap partner of the top quark, which appears in the core with degree 2 and is the observer partner in the Klein orbit structure of HCSM-27.

5.3 Relation to the Klein orbit structure

The four isolated edges of the distinguished-pair graph correspond to four of the six Klein orbits from HCSM-27: specifically, the four two-cycle orbits with pointwise stabilizer generated by the product of the two Klein generators. The remaining two Klein orbits — the observer orbit $\{W, t\}$ and the four-cycle orbit $\{\text{dark}, s, u, \tau\}$ — are the two halves of the six-vertex core, joined by the bridge edge $\{\text{dark}, t\}$.

The distinguished-pair graph therefore provides a natural presentation of the Klein orbit structure as a graph.

6 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at the verification folder with the number twenty-eight. The script constructs the fourteen-mode multiplet, the dihedral orbit partition, the winding class partition, the fermion/boson split, the swap pair, and the distinguished-pair graph from HCSM-native definitions, and verifies:

1. The twelve modes on which the dihedral orbit partition and the winding class partition agree.
2. The two modes on which the partitions differ: the top quark and the dark photon.
3. The unique two-mode swap that makes the winding classes fermion/boson-homogeneous.
4. The ten edges of the distinguished-pair graph.
5. The decomposition into four isolated edges and one connected six-vertex core.
6. The homogeneity of each isolated edge in dihedral orbit, winding class, and fermion/boson content.
7. The heterogeneity of the six-vertex core in every structural feature.
8. The unique three-degree hub at the dark photon.
9. The cycle rank 1 of the six-vertex core and the explicit cycle $\text{dark} \rightarrow u \rightarrow s \rightarrow \tau \rightarrow \text{dark}$.

Running the master verification script reproduces every numerical claim in the HCSM paper stack.

7 Conclusion

This paper has proven two structural theorems about the fourteen-mode multiplet of the Hodge Complex Standard Model.

The first is the Orbit-Class Reconstruction Theorem: the winding class partition is the dihedral orbit partition modified by a unique two-mode swap consisting of the top quark and the dark photon. The swap is uniquely forced by requiring each winding class to be fermion/boson-homogeneous. Equivalently, the winding class partition is not an independent structure but a minimal correction to the dihedral orbit partition.

The second is the Distinguished-Pair Graph Decomposition Theorem: the graph of the ten distinguished pairs of the fourteen-mode multiplet decomposes into four isolated two-vertex edges and one connected six-vertex core. Every isolated edge is homogeneous in dihedral orbit, winding class, and fermion/boson content, and heterogeneous in charge-line sign. The six-vertex core is heterogeneous in every structural feature, has a unique three-degree hub at the dark photon, and contains exactly one independent cycle.

Together, the two theorems reveal that the fourteen-mode multiplet has a two-tier pair structure: an outer shell of four homogeneous edges, and an inner core that spans every partition of the multiplet and contains the framework's unique pair bridge between the observer pair and the four-cycle orbit.

References

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A Web Crawler Metadata

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 the Orbit-Class Reconstruction Theorem: the winding class partition
 is the dihedral orbit partition modified by a unique two-mode swap
 consisting of the top quark and the dark photon, uniquely forced by
 requiring each winding class to be fermion/boson-homogeneous. Second,
 the Distinguished-Pair Graph Decomposition Theorem: the graph of the
 ten distinguished pairs decomposes into four isolated two-vertex edges
 and one connected six-vertex core; every isolated edge is homogeneous
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 six-vertex core is heterogeneous in every structural feature, with a
 unique three-degree hub at the dark photon and cycle rank one.
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A.2 Schema.org JSON-LD

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    requiring each winding class to be fermion/boson-homogeneous. Second,
    the Distinguished-Pair Graph Decomposition Theorem: the graph of the
    ten distinguished pairs decomposes into four isolated two-vertex edges
    and one connected six-vertex core; every isolated edge is homogeneous
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    "Hodge Complex Standard Model",
    "HCSM",
    "orbit-class reconstruction",
    "distinguished-pair graph",
    "winding class partition",
    "dihedral orbit partition",
    "fermion/boson split",
    "swap pair",
    "cycle rank",
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  "license": "All rights reserved",
  "identifier": "HCSM-28-2026-09",
  "isPartOf": {
    "@type": "Periodical",
    "name": "HCSM White Paper Series"
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A.3 BibTeX for Automated Indexing

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@article{PreschuttiHCSM28,
  author = {Preschutti, Stanley},
  title = {{HCSM-28}: Orbit-Class Reconstruction},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, orbit-class reconstruction,
    distinguished-pair graph, winding class partition,
    dihedral orbit partition, fermion/boson split, swap pair,
    cycle rank, graph decomposition},
  abstract = {This paper proves two structural theorems about the
    fourteen-mode multiplet of the Hodge Complex Standard Model. The
    Orbit-Class Reconstruction Theorem identifies the winding class
    partition as the dihedral orbit partition modified by a unique
    two-mode swap. The Distinguished-Pair Graph Decomposition Theorem
    decomposes the graph of the ten distinguished pairs into four
    isolated edges and one connected six-vertex core.}
}
```